

TOI-4087b

R-2025

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_260329 · created 2026-08-08T00:37:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461128.834 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.04
Transit depth (Rp/Rs)^2	1.79 +/- 1.06 [%]
Orbital Inclination (inc)	85.46 +/- 2.07
Airmass coefficient 1 (a1)	0.0076 +/- 0.0043
Airmass coefficient 2 (a2)	2.92 +/- 1.6
Scatter in the residuals of the lightcurve fit is	53.63 %
Best Comparison Star	#2 - [221, 385]
Optimal Aperture	6.22
Optimal Annulus	16.47
Transit Duration (day)	0.062 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.134 ± 0.04 vs 0.10754068997360952 (deviation 0.65σ)
Duration fit vs published	0.062 d vs 0.1242 d (deviation 2.54σ)
Mid-time O–C	31.3 min
Depth SNR	3.3
Residual scatter	53.63 %

Light curve

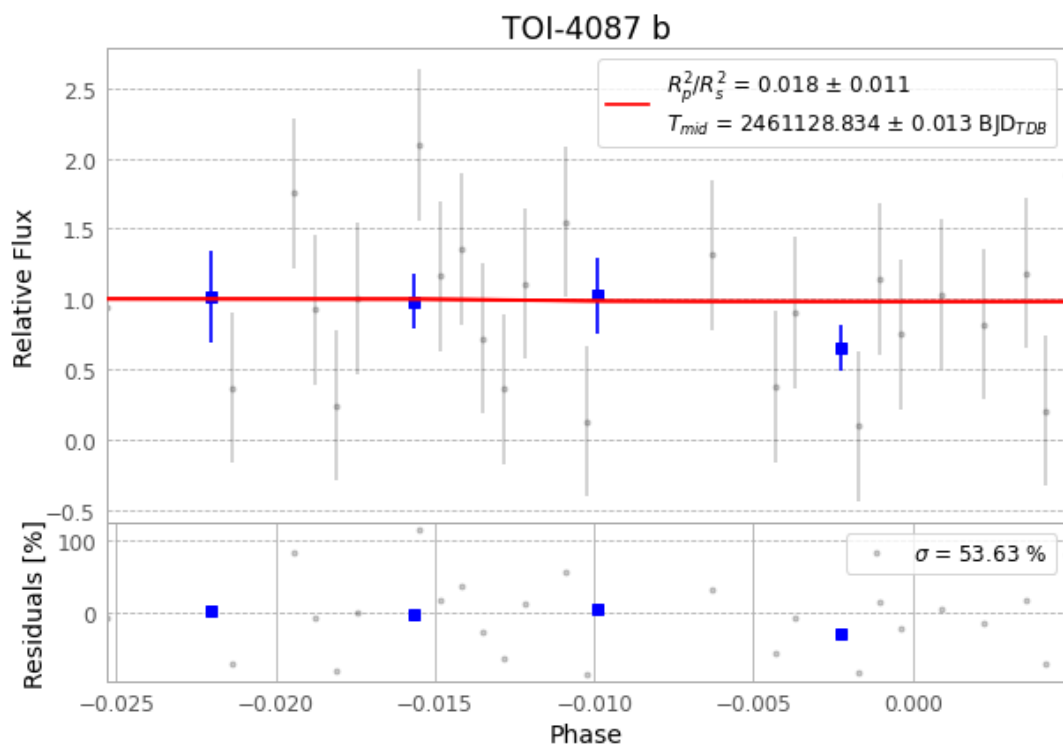


Figure 1 — FinalLightCurve_TOI-4087 b_2026-03-28.png (SHA-256 ee4790454a8f1805...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [434, 177], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e04d26c33569c6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

53 FITS frames (35 MB), TOI4087260329054515.FITS → TOI4087260329082116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-260329.log (4f1644ad39e4...), FinalLightCurve_TOI-4087 b_2026-03-28.png (ee4790454a8f...), FinalLightCurve_TOI-4087 b_2026-03-28.csv (feca6246d625...)

Compute resources

Wall time 158.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:37Z.